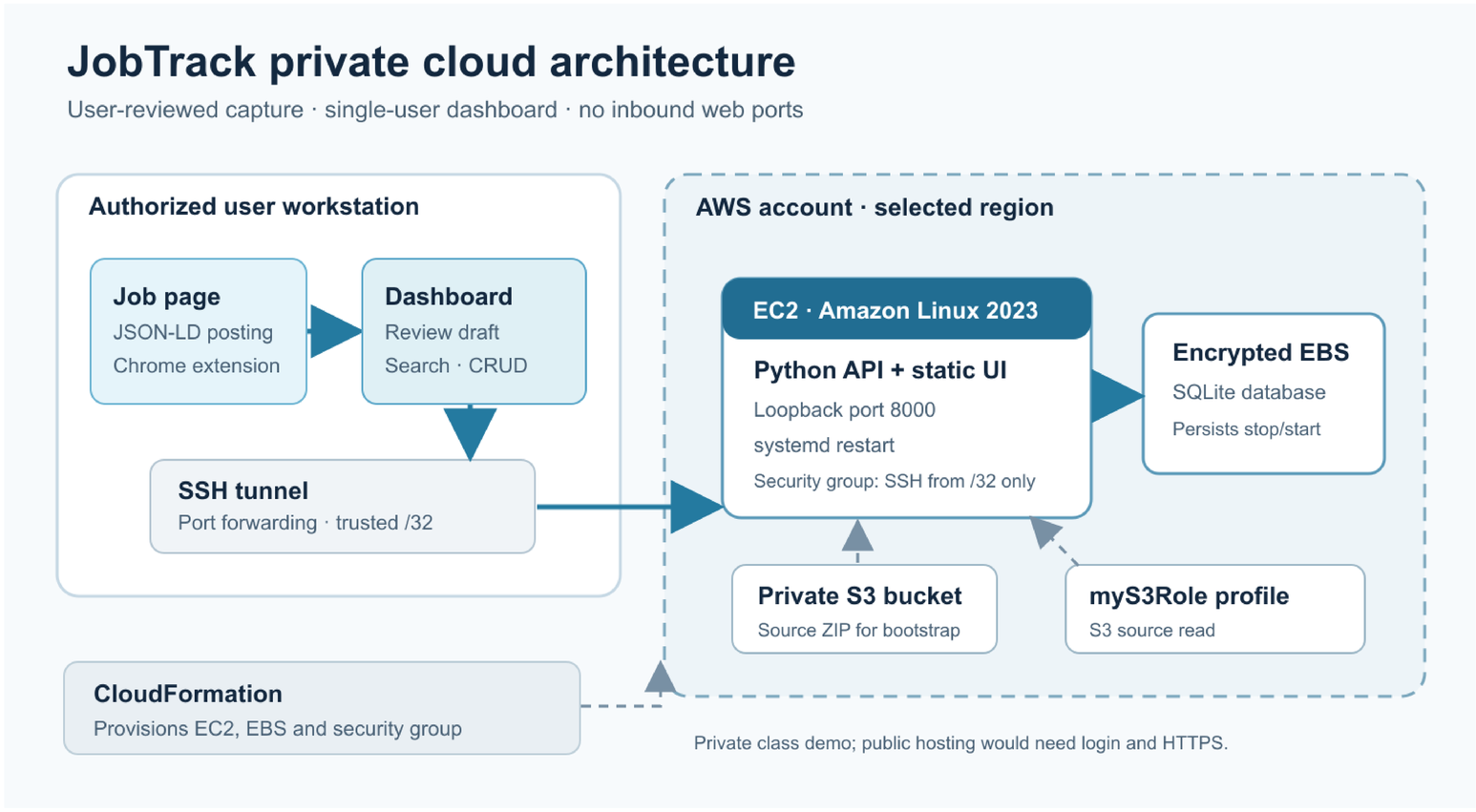
JobTrack

A private cloud-hosted job application tracker

CSE 4333 Cloud Computing · Group 42

Ahmad Salem · NoorAldeen Waleed



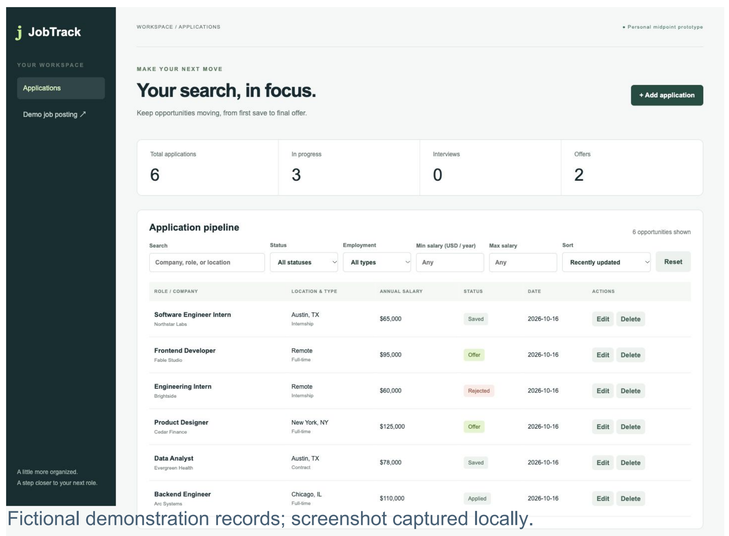
Project summary

JobTrack keeps a student's job applications in one searchable pipeline. It stores company, role, location, annual salary, status, date, link, and notes.

The dashboard, Python API, and SQLite support tracking and job discovery. Remotive works without a key; Indeed and LinkedIn use optional JSearch. A Chrome extension reads structured postings.

Verified private AWS sandbox deployment: EC2, encrypted EBS, private S3 source delivery, IAM, CloudFormation, and a restricted SSH tunnel.

Application dashboard



Fictional demonstration records; screenshot captured locally.

Introduction · users

Students and recent graduates often manage many applications at once and lose track of dates and next steps.

JobTrack keeps opportunities, status changes, salary, and notes together so a user can see their pipeline and act on it.

User workflow

Find a listing or import a posting draft; review fields; save; update status; search and filter; delete old entries.

Remotive and optional JSearch provide job discovery. The extension reads Schema.org JobPosting JSON-LD. JobTrack does not apply to employers.

Conclusion

JobTrack puts discovered opportunities and application progress in one searchable place. It is a working private single-user demonstration.

AWS deployment and performance are described using measured results; public multi-user hosting remains future work.

Application logic

The responsive dashboard calls same-origin JSON endpoints. The Python API validates fields and stores UTC timestamps in SQLite.

Endpoints cover health, application CRUD, /api/jobs discovery, /api/job-sources status, and saving a listing. Search and filters combine; sort keys are allowlisted.

Cloud services

CloudFormation provisions EC2, encrypted EBS, and a security group allowing SSH only from one trusted /32. The sandbox's myS3Role profile pulls source from private S3.

An SSH tunnel lets an authorized user reach the loopback-only app without opening a public web port.

Alternatives considered

A public HTTPS app needs login and per-user authorization. RDS or DynamoDB would improve managed storage; serverless hosting could scale differently.

The one-instance SQLite design is appropriate for a small private class demo, but it has no high availability or automatic backup.

Challenges and lessons

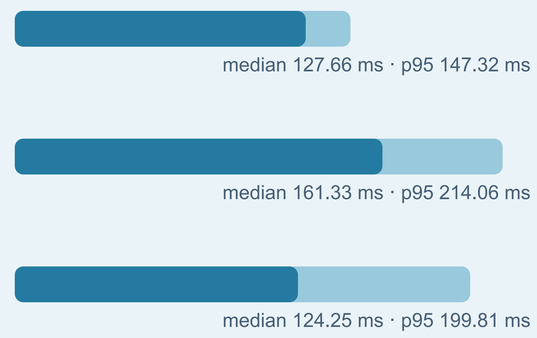
Browser QA exposed a Reset-button conflict. Final API review exposed unclosed SQLite connections. Both were fixed and regression-tested.

A restricted SSH tunnel keeps the web service private. The sandbox session is time-limited; the EBS root volume needs a backup before termination.

Team contributions

Ahmad Salem and NoorAldeen Waleed — equal co-developers. NoorAldeen built the core tracker and job-discovery features. Ahmad handled final cloud deployment, testing, documentation, poster, and slides. Both shaped the project design.

Performance evaluation



50 sequential GET requests per endpoint after 5 warmups; 5 fictional records. t3.micro, us-east-1; includes the client-to-EC2 SSH tunnel. Small single-user sample.

Verification

23 Python tests and 5 extractor tests pass. The browser checked draft review, save, edit, refresh, filters, sorting, deletion, and job discovery.

Cloud CRUD, restart persistence, and Remotive discovery passed through the SSH tunnel. The installed extension toolbar was not tested.

Machine learning

Not applicable. JobTrack uses rules and a relational database; it has no trained model.

Future work

Add authentication and user isolation before public access. Move to managed backup/database, add HTTPS and monitoring, and broaden posting extraction.

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